

Observing Case: Does the Galactic Acceleration Scale Evolve?

Deep-Regime Rotators at $z \simeq 2\text{--}2.6$

Version 2: corrected target ledger, separated rivals, honest odds

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Abstract

Companion to the pre-registered measurement (doi:10.5281/zenodo.22563139), written for observers. This version supersedes version 1, whose candidate ledger carried values that contradict their own cited sources, and which understated what a single galaxy can decide. Three readings of the radial-acceleration scale a_0 make different zero-parameter predictions for its value at redshift z , quoted separately here as $\Delta \equiv \log_{10}[a_0(z)/a_0(0)]$ at $z = 2.5$: constant (0.00), the emergent scale of Λ CDM haloes (+0.23 to +0.46 dex over halo masses $10^{11}\text{--}10^{13} M_\odot$ and two concentration–mass relations), and $a_0 \propto H(z)$ (+0.58 dex). With intrinsic and halo-to-halo scatter included, one galaxy measured to ± 0.13 dex gives odds of about 4:1 between constant and Λ CDM-emergent, not 20:1. That decision needs three galaxies at ± 0.10 dex, or four at ± 0.20 . After every row of the ledger was re-read from its primary paper, no known galaxy passes the gate. One rotation-supported, low-acceleration galaxy at $z \geq 2$ is a viable first target: OLAS M0717-02064 at $z = 2.07$. Its published kinematics favour none of the three readings. For velocity data alone, the per-object precision is gas-limited at ≈ 0.43 dex. A gas mass good to ± 0.12 dex brings it to ≈ 0.26 dex.

What changed from version 1

1. **The ledger was wrong.** Several rows contradicted their cited sources. The replacement, `highz_target_ledger_verified_2026.py`, re-reads every value from its primary paper. The largest error: A68-HLS115 lies at $z = 1.5869$ with $M_\star = 8.1 \times 10^9 M_\odot$ (Dessauges-Zavadsky et al. 2015). It is not at $z = 2.49$, and it is not a deep-regime candidate.
2. **Two rivals were merged.** Version 1 called the Λ CDM-emergent rise “ $\propto H(z)$ ”. The emergent law, $E(z)^{4/3}c^2/f(c)$, gives +0.33 dex at $z = 2.5$ for $10^{12} M_\odot$ haloes with Dutton–Macciò concentrations. The $H(z)$ law gives +0.58 dex. They are now separate hypotheses.
3. **One galaxy does not decide.** The “20:1 with one galaxy at ± 0.13 dex” ignored the relation’s intrinsic scatter and the halo-to-halo concentration scatter. The corrected odds are in Sec. 3.
4. **Sign convention.** Version 1 defined $\Delta_{\text{BTFR}} = \log M_b - 4 \log V_f - C_0$ but quoted a rise as positive. For a rising a_0 that quantity is negative. This version uses $\Delta = \log_{10}[a_0(z)/a_0(0)]$ throughout.

5. **A caveat on the framework’s own prediction.** The constant law is the framework’s, but its observable zero point is not yet sharp. A retained cold halo around $z \simeq 2.5$ galaxies (internal note L189) raises the apparent, baryons-only zero point by $\approx +0.2$ dex in the one example worked there, a $3 \times 10^{11} M_\odot$ halo around a rotator with $10^{10} M_\odot$ of baryons. That shift points toward the lower end of the Λ CDM range. The correction has not been computed. Until it is, this case tests the three hypotheses above and not the framework as a whole.

1 The three hypotheses

$\Delta = \log_{10}[a_0(z)/a_0(0)]$	constant	Λ CDM emergent	$a_0 \propto H(z)$
$z = 2.07$ (target 1)	0.00	+0.17 to +0.26	+0.50
$z = 2.5$	0.00	+0.23 to +0.46	+0.58
$z = 2.62$ (target 2)	0.00	+0.24 to +0.35	+0.60

At target redshifts the Λ CDM ranges are for 10^{11} – $10^{12} M_\odot$ haloes (DM14). At $z = 2.5$ the range spans 10^{11} – $10^{13} M_\odot$ and both the DM14 and D08 concentration–mass relations. In the deep regime, $V_c \propto a_0^{1/4}$ at fixed baryonic mass. The local calibration is frozen before any high-redshift value is examined, and both local footings (9.36×10^{-11} and $1.13 \times 10^{-10} \text{ m s}^{-2}$) are always reported.

2 The gate (unchanged except the window)

A galaxy enters the decisive sample only if all of the following hold:

1. $g_{\text{bar}}(R_{\text{out}}) < 0.3 a_0$ at the upper end of its mass range;
2. $V_{\text{rot}}/\sigma > 1.5$ in a forward-modelled source-plane fit, with inclination, dispersion, beam smearing and asymmetric drift fitted together;
3. $\delta \log \mu \lesssim 0.05$ dex (tightened from 0.08: the magnification enters the inverted a_0 with weight $A_{\text{obs}}/2 \approx 1.3$);
4. $35^\circ < i < 75^\circ$, with at least 5 source-plane resolution elements along the major axis.

The redshift window is relaxed from $2.32 < z < 3.12$ to $z \geq 2$: at $z = 2.07$, $\text{H}\alpha$ falls in NIRSspec G235H, $[\text{O III}]$ and $\text{H}\beta$ fall in G140H, and CO(3–2) falls at 112.6 GHz, inside ALMA Band 3.

3 What one, two, three galaxies decide

Near the gate the inversion $a_0 = g_{\text{bar}}/y$ of the relation $\nu(y) = 1/(1 - e^{-\sqrt{y}})$ amplifies errors. At $y = 0.2$, a mass error is amplified $\times 1.52$ and an error in g_{obs} (twice the log-velocity error) $\times 2.52$. The deep-limit estimator $4 \log V - \log M_b$ understates both. Two scatters add to the measurement error: the relation’s intrinsic scatter (0.034 dex in g_{obs} , i.e. 0.08–0.09 dex in a_0), and, on the Λ CDM side only, ± 0.13 dex in each object’s expectation from halo-to-halo concentration scatter. Expected odds between constant and Λ CDM-emergent ($\Delta = 0.33$) at $z = 2.5$:

σ_{meas} per galaxy	$N = 1$	$N = 2$	$N = 3$	$N = 4$
0.10 dex	5.0:1	24.6:1	122:1	606:1
0.13 dex	3.8:1	14.1:1	53:1	199:1
0.20 dex	2.3:1	5.4:1	12.5:1	28.9:1

Velocity data alone (gas from the resolved star-formation law, ± 0.30 dex; V_c to 5%; inclination $\pm 8^\circ$): $\sigma(\log a_0) \approx 0.43$ dex per object at $y = 0.2$. Three such objects separate constant from $a_0 \propto H(z)$ at $\approx 2\sigma$ and do not separate constant from Λ CDM-emergent ($\approx 0.7\sigma$). **With a gas mass good to ± 0.12 dex** (CO or dust continuum, with metallicity-dependent conversion), the error is ≈ 0.26 dex per object. Three objects then reach $\approx 3.5\sigma$ on constant versus $H(z)$. Constant versus Λ CDM-emergent still needs the tighter budgets in the table. *The gas mass is the hard half of the measurement.*

4 Candidates (verified ledger)

OLAS M0717-02064 ($z = 2.07$, $\mu = 6.48$, RA/Dec 07:17:39.125 +37:44:18.45; Hirtenstein et al. 2019, Tables 1–2) is lens-corrected $\log M_\star = 8.08 \pm 0.06$, with $\Delta v = 142 \text{ km s}^{-1}$, $\sigma_{\text{local}} = 42.9 \text{ km s}^{-1}$, $v/\sigma = 1.64$ (class “rotation”) and $\text{H}\alpha$ SFR $4.5 M_\odot \text{ yr}^{-1}$. It is the only $z \geq 2$ object that is both rotation-supported and plausibly deep-regime, and it has no JWST spectroscopy yet (the MAST archive, 25 September 2026, shows NIRCcam imaging only). *Face value.* Invert the full relation at $R = 1.5\text{--}3.5$ kpc on both footings. A constant scale needs $M_{\text{gas}}/M_\star \approx 6\text{--}36$ (depletion time 0.17–0.95 Gyr), and $a_0 \propto H(z)$ needs $\approx 3\text{--}17$. Both are plausible, so **the published data favour none of the hypotheses**. (An intermediate calculation with the deep-limit formula suggested a lean against constancy. It overstated the required gas by $\times 1.5\text{--}3.4$ and is withdrawn.) There is also a risk: if the disc is inclined and the scale is constant, the galaxy sits at $y \approx 0.4\text{--}1.8$ within 3.5 kpc and could fail gate 1 unless the velocity field reaches further out.

Abell 68 C4 ($z = 2.622$, $\mu \approx 35\text{--}46$) has two published stellar masses 0.7 dex apart ($\log M_\star = 8.7$, Richard et al. 2007; 8.0, Richard et al. 2011). It is deep-regime-plausible only at the lower value. No kinematics exist. **Abell 68 C20b** ($\mu \approx 158$, range 83–302) fails the lens gate. **SL2S J0217** ($z = 1.844$, $\mu = 17.3$) has $\log M_\star = 8.26$ and a metallicity of $0.05 Z_\odot$ (Berg et al. 2018). Existing ALMA [C II] (tentative) and PdBI CO(2–1) (undetected) data constrain its cold gas (Rybak et al. 2021), but its CO is expected to be dark. **Controls:** A1689B11 ($z = 2.54$; $\text{H}\alpha$ detected only to ≈ 1.7 kpc; $V/\sigma = 9\text{--}13$ in Yuan et al. 2017) and zC-400569 ($z = 2.24$; $V_{\text{rot}} = 254 \pm 41 \text{ km s}^{-1}$, $V/\sigma \gtrsim 17$), both high-acceleration. They must return the local zero point, where all three hypotheses agree.

5 Decision rule

The rule is stated per hypothesis and is unchanged in form. Measured Δ near 0.00 keeps the constant law and excludes the others at the achieved odds. Near the Λ CDM range, constancy is disfavoured. Near $+0.5\text{--}0.6$, both constancy and the emergent law are disfavoured. A value inconsistent with all three is reported as such. Odds are quoted with the scatters of Sec. 3, never from the measurement error alone. A null on the gate (no galaxy qualifies) is itself informative about the deep-regime population at $z \geq 2$.

Provenance. doi:10.5281/zenodo.22563139 (pre-registration). The repository scripts are `prep_2026/a0z_crossscale/highz_target_ledger_verified_2026.py` (verified ledger) and `prep_2026/a0z_crossscale/a0z_deepmond_ifu_forecast_2026.py` (hypotheses, face value, per-object precision; 14/14 checks). The odds table uses the same scatter model as the programme's step-by-step manuscript (`paper_numbers.py`, S4).